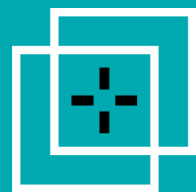




ceci n'est pas une conférence

Supporting Mobility Decisions with MapStore



GEO SOLUTIONS
unleashing location data

Yanko Godaert



OUR CLIENTS





What is MapStore?





MapStore

- **Free & Open Source** WebGIS software
- **GeoSolutions** (☒) product owner MapStore (& GeoNode, GeoServer, GeoNetwork core contributor)
- **Geoportal** for creating, managing and sharing **maps, dashboards, geostories**
- **'out-of-the-box'** lot of default GIS-viewer-functionalities
- **Advanced functionalities** (widgets, timeline, dashboards, geostories, 3D, ...)
- **Adaptive** (desktop - mobile)
- Highly **customizable**
 - Open Source
 - Extension framework
 - Downstream projects

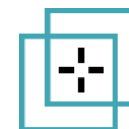


MapStore@DMOW



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Department of Mobility and Public Works (DMOW)



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- Governmental institution of Flemish government (Flanders region)
- Make Flanders move along
 - 85 000 km roads
 - 1 000 km waterways
 - 4 ports

>> *big challenges in **safe, smart and sustainable mobility policy making with adequate infrastructure***

- Highly-location related domain >> *benefit from **geo-data enabled decision-making***
 - MapStore
 - GeoServer
 - GeoNetwork





MapStore@DMOW



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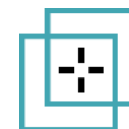
- **4 y setup open-source web-GIS-infra**
 - GIS-viewer: **MapStore** downstream project
 - Webservices: **GeoServer**
 - Metadata portal: **GeoNetwork**
 - Databases: PostgreSQL
- **Maps, dashboards & geostories** for different use cases
 - Internal
 - Public
 - Maps & geostories
 - Embedded
- **Support contract GeoSolutions (Italy):** feature requests, priority bug fixes & configuration support for MapStore, GeoServer, GeoNetwork
- **Role Geo Solutions (Belgium)**
 - Not involved with initial setup (other projects)
 - Manage & support MapStore project
 - E&E data processing



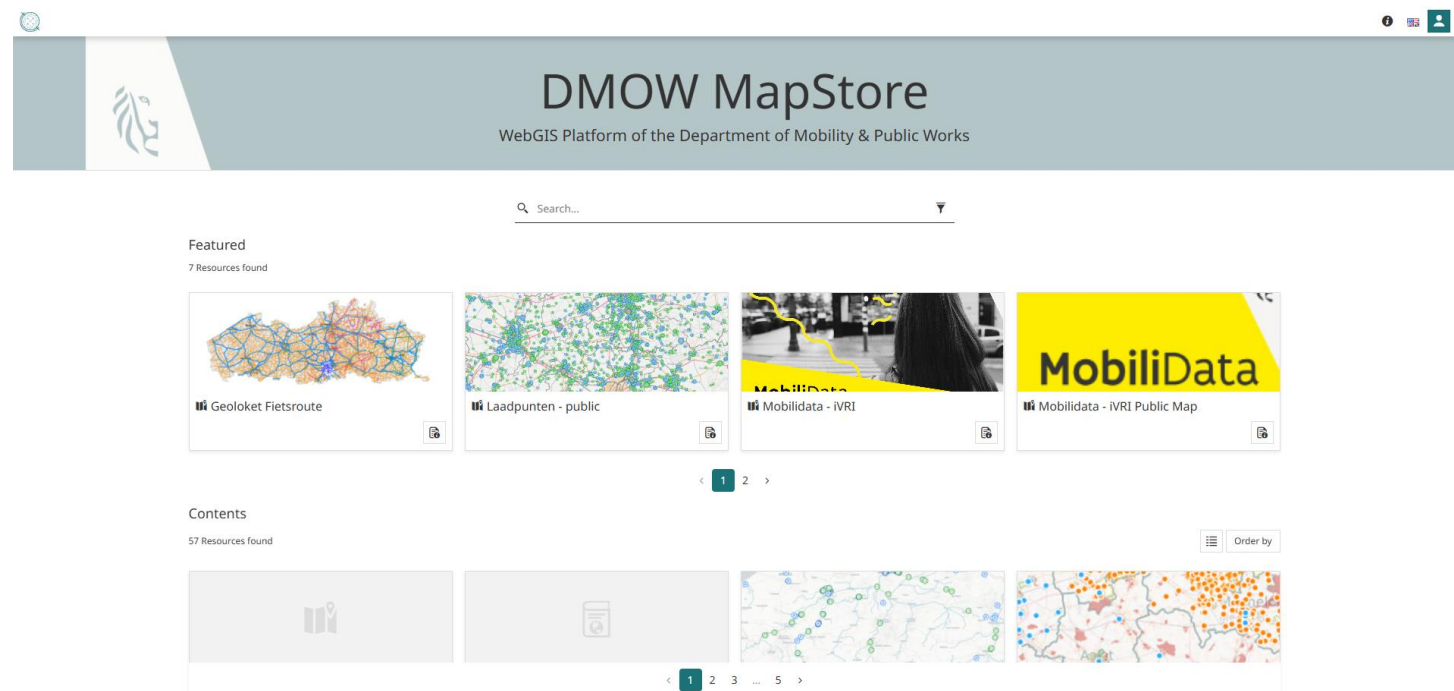
DMOW GIS Viewer



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Case: Shared soft mobility

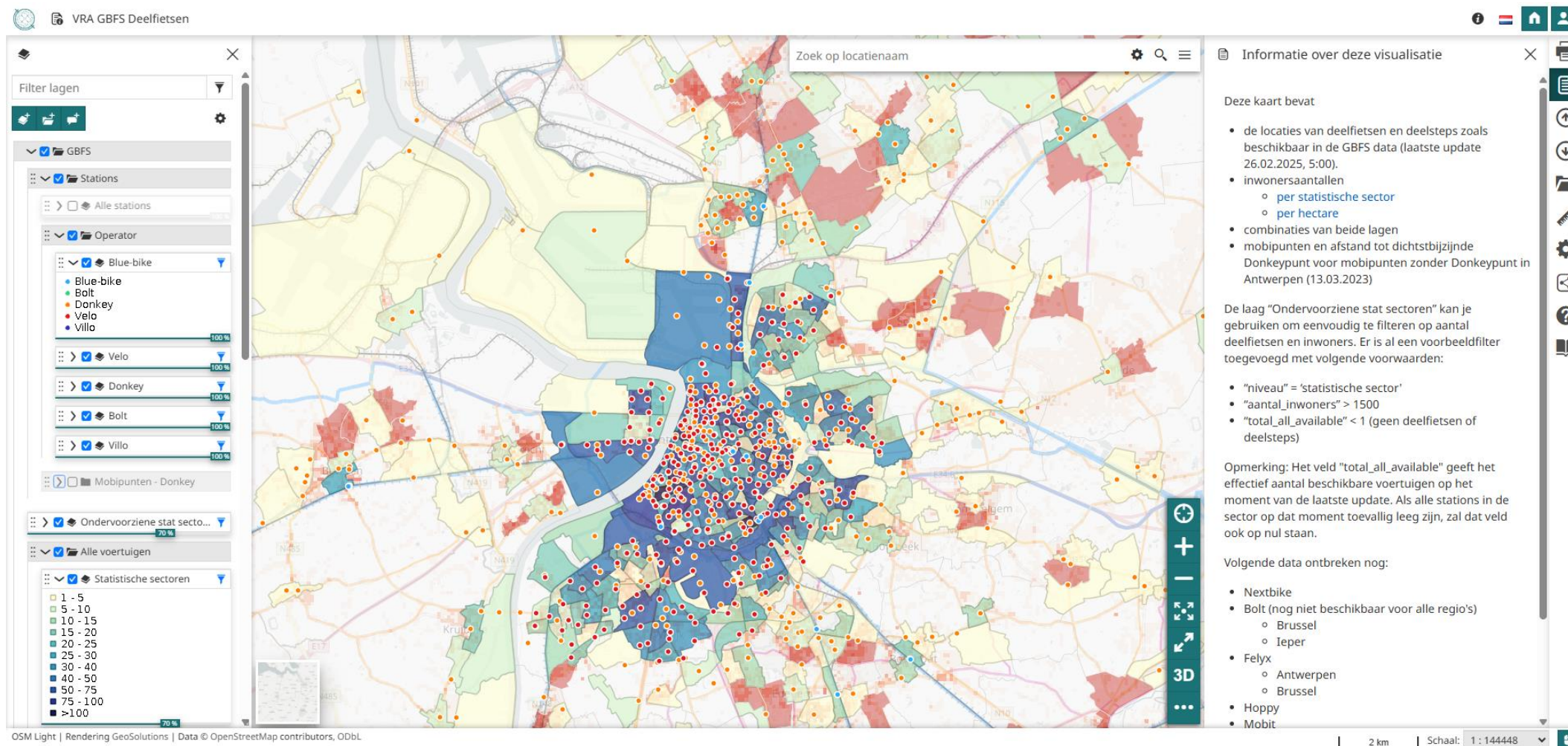
- Transport region Antwerp:
 - **Current situation of shared soft mobility** (bikes & scooters)?
 - **Where can we improve?** Which areas are underserved?
- **Processing**
 - harvest all available **open [GBFS](#) data**
 - aggregate on different levels of **administrative boundaries**
 - enrich with number of **inhabitants**
 - entire **Flanders region**
- Publish on **GeoServer**, with multiple styles
- Create public **MapStore map** >> from open data to open data



Demo: Shared soft mobility



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Case: Intelligent Traffic Lights



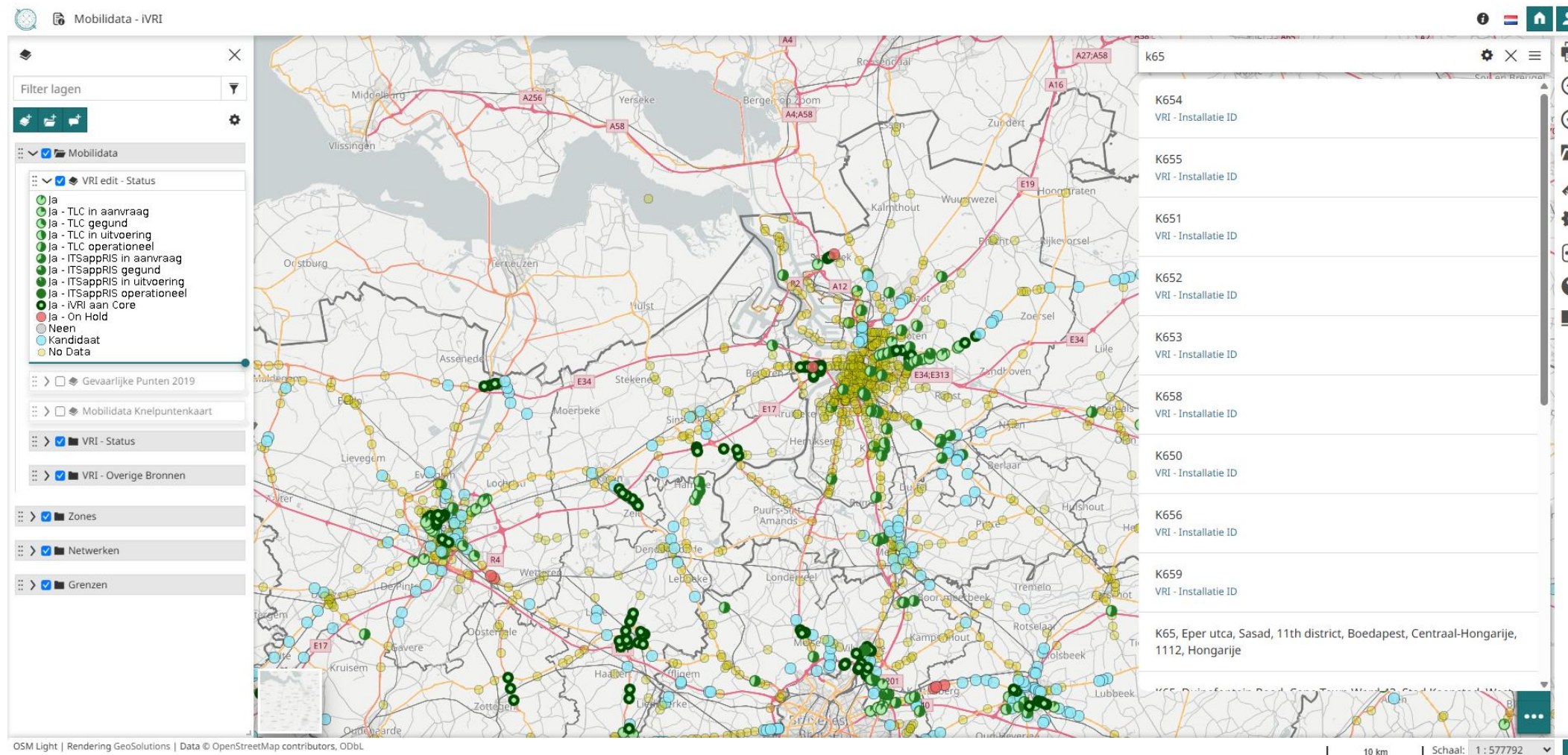
- Equip existing traffic lights with new **smart technology**
 - **React smoothly and adaptively** to road users at an intersection
 - **Turning green faster or longer** based on the actual traffic ahead
 - **Two-way communication** (e.g., emergency services)
- **Current situation**
 - Active: 205
 - Under construction: 149
 - Scheduled: 63
- **MapStore goals**
 - Manage data & status >> **WFS transaction** editing capability
 - Communication to **citizens**
 - Simplified public map view
 - Embedded on website



Demo: Intelligent Traffic Lights



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Sidenote: Dutch translations



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- **MapStore UI** configured in Dutch
- On every **MapStore update** DMOW
 - Checks & updates Dutch translations
 - Pushes translations to source code





Sidenote: Authentication



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1. Default internal authentication service

2. LDAP/AD service

3. OpenID service

- OpenID Connect
 - Default providers
 - Google
 - Keycloak
 - Custom **ACM/IDM** provider
 - <https://authenticatie.vlaanderen.be>
 - Login VO/Itsme
 - User management not in MapStore but external in ACM/IDM
- >> [generic \(experimental\) OpenID Connect provider](#)



Case: PhotoViewer extension

- Custom MapStore extension >> developed from **extension framework**
- **Functionalities**
 - View a picture on a map, linked to attributes in a dataset
 - Implement routes
 - Navigation
 - Animation
 - Generic >> lot of use cases
 - Custom street view
 - Graphs of FCD travel times
 - **Historical aerial photographs >> demo POC**



Demo: PhotoViewer extension



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ATO_Tielt_11_60891

<< < > >>

Trefwoord_Tooniem : Gent - Drongen - Baarle

Kaartblad : 21/4/6b - 21/1/5b

Datum : 1961-08-29Z

Uur : 12.47

Hoogte : 1160

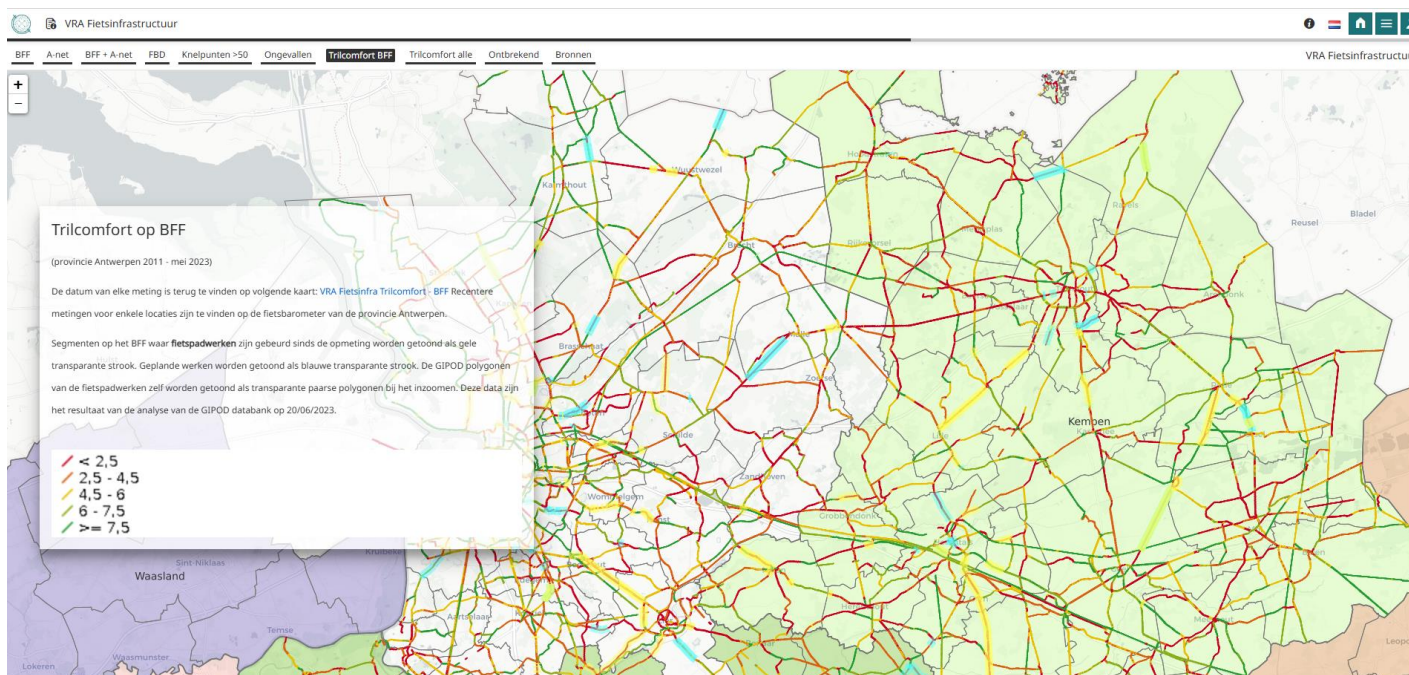
Brandpuntafstand : F.11,4



Case: Geostory bicycle infrastructure



- Transport region Antwerp
- Geostory to integrate a lot of complex location data in different maps into **one thematic overview** with extra contextual information in a Geostory
- Integrated in reports >> **publicly shared with citizens**





THANKS FOR YOUR ATTENTION.

For more information, visit www.geosolutions.be.